CHAPTER

18

FFT Convolution

This chapter presents two important DSP techniques, the *overlap-add method*, and *FFT convolution*. The overlap-add method is used to break long signals into smaller segments for easier processing. FFT convolution uses the overlap-add method together with the Fast Fourier Transform, allowing signals to be convolved by multiplying their frequency spectra. For filter kernels longer than about 64 points, FFT convolution is faster than standard convolution, while producing exactly the same result.

The Overlap-Add Method

There are many DSP applications where a long signal must be filtered in *segments*. For instance, high fidelity digital *audio* requires a data rate of about 5 Mbytes/min, while digital *video* requires about 500 Mbytes/min. With data rates this high, it is common for computers to have insufficient memory to simultaneously hold the entire signal to be processed. There are also systems that process segment-by-segment because they operate in *real time*. For example, telephone signals cannot be delayed by more than a few hundred milliseconds, limiting the amount of data that are available for processing at any one instant. In still other applications, the *processing* may require that the signal be segmented. An example is FFT convolution, the main topic of this chapter.

The overlap-add method is based on the fundamental technique in DSP: (1) decompose the signal into simple components, (2) process each of the components in some useful way, and (3) recombine the processed components into the final signal. Figure 18-1 shows an example of how this is done for the overlap-add method. Figure (a) is the signal to be filtered, while (b) shows the filter kernel to be used, a windowed-sinc low-pass filter. Jumping to the bottom of the figure, (i) shows the filtered signal, a smoothed version of (a). The key to this method is how the *lengths* of these signals are affected by the convolution. When an N sample signal is convolved with an M sample